Mg-Fe Layered Double Hydroxides as Precursors for Ammonia Decomposition Catalysts

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Introduction

Ammonia is widely recognized as a promising and sustainable hydrogen carrier for efficient hydrogen storage and transport.[1] For its catalytic decomposition to release hydrogen iron-based catalysts are often employed due to their high activity and cost-effectiveness. However, these catalysts are prone to deactivation due to the formation of bulk iron nitrides.[2,3] To better understand this phenomenon, we aim to elucidate the deactivation mechanisms and establish a structure-activity relationship using operando characterization techniques. A novel method will be established where neutron scattering is combined with a compact profile reactor to gain comprehensive insights into the mechanism. This knowledge will enable the rational design of resilient iron-based catalysts.

Experimental

A robust model catalyst is synthesized from magnesium-iron layered double hydroxide with the formula $Mg_4Fe_2(OH)_{12}CO_3{\cdot}3H_2O$, leveraging cost-effective and abundant precursors.[4] It can be prepared using the constant pH co-precipitation method leading to crystalline precursors with a mixed cationic lattice containing all metal species of the resulting catalyst in a single-source precursor.[5] The synthesis is scalable, reproducible and adaptable, allowing for the incorporation of various metal ions to enhance catalyst performance by suppression of the nitride formation based on insights from simultaneous structural analysis and catalytic testing. First, we have studied the effect of calcination temperature leading to different precatalyst materials. Upon decomposition, this precursor yields high surface area oxides with phase compositions ranging from amorphous forms to α-Fe_2O_3-MgO mixtures and inverted spinels, controllable through the calcination temperature. The thermo-gravimetric analysis of the precursor in air with a heating rate of 1 $Kmin^{-1}$ is shown in Fig. 1a) and depicts four mass loss steps yielding different precatalysts for catalytic testing. The

effect of different calcination temperatures and their relation to the crystallographic phase composition was investigated with X-ray Powder diffraction (Fig. 1b).

The catalytic testing will be complemented by the characterization of the nitride evolution of the different precatalysts in ammonia. This approach enables a direct comparison of different crystallographic phases and their structure-activity relationship, all while using the same precursor to maintain identical metal ratios.

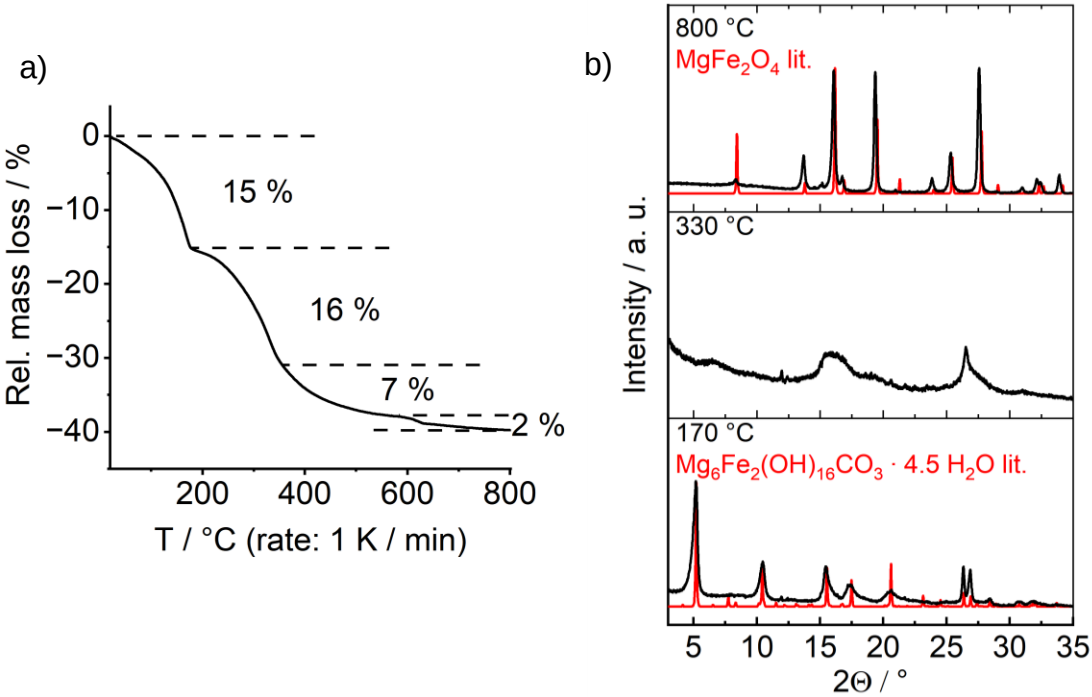

Fig. 1: Results of the thermo-gravimetric analysis is shown on the left side (a) and X-ray powder diffraction from precursors calcined at different temperatures on the right side (b).[6,7]

[1] S. Ristig, M. Poschmann, J. Folke, O. Gómez-Cápiro, Z. Chen, N. Sanchez-Bastardo, R. Schlögl, S. Heumann, H. Ruland, *Chem. Ing. Tech.*, 94, **2022**, 1413-1425.

[2] S. Chen, J. Jelic, D. Rein, S. Najafishirtari, F.-P. Schmidt, F. Girgsdies, L. Kang, A. Wandzilak, A. Rabe, D. E. Doronkin, J. Wang, K. F. Ortega, S. DeBeer, J.-D. Grunwaldt, R. Schlögl, T. Lunkenbein, F. Studt, M. Behrens, *Nat. Commun.*, 15, **2024**, 871.

[3] B. Lu, L. Li, M. Ren, Y. Liu, Y. Zhang, X. Xu, X. Wang, H. Qui, *Appl. Catal. B: Enviromental*, 314, **2022**, 121475.

[4] F. Kovanda, V. Balek, V. Dornicak, P. Martinec, M. Maslan, L. Bilkova, D. Kolousek, I. M. Bountsewa, *J. Therm. Anal. Calorim.,* 71, **2003**, 727-737.

[5] M. Behrens, Catal. Today, 246, **2015**, 46-54.

[6] A. Olowe, *AXA*, 38, **1995**, 749-755.

[7] L. Passerini, *Gazz. Chim. Ital.*, 60, **1930**, 389-399.